Kinship Screening with QAOA: 30-Day Hackathon Plan

Nov 12, 2026 · @meow2

Summary

The project is a planning tool for thalassemia cascade screening: for one home visit with k test kits, it chooses which relatives to test. Its objective is the expected number of at-risk couples found. That value sits on pairs of people, so it maps directly onto QAOA. The plan covers 30 days, 12 Nov to 11 Dec 2026, and ends with all four required deliverables submitted a day early.

Three things decide the result:

- **The problem must be non-trivial.** If "test the closest relatives first" is always the best choice, there is nothing to optimise. Day 10 is a go/no-go gate on this.
- **The quantum claim must be honest.** We do not claim classical computers cannot solve it. We claim a natural mapping, a hard problem class and a measured comparison.
- **Every deliverable must exist and open.** A missing video or a dead link costs more than any technical weakness.

What the brief demands

The brief sets six steps and four submission items, and the plan covers each one by name.

Deadline check. The brief you shared states 12 Nov 2026, 11:59 pm. This plan assumes the real deadline is about a month later and targets submission on `Dec 11, 2026`. Confirm the date with the organisers on Day 1.

Step in the brief	What it asks for	Our output
1. Identify	A concrete, relevant optimisation problem and its users	One-page problem statement; users are screening field officers and programme managers

Step in the brief	What it asks for	Our output
2. Formulate	Decision variables, objective, constraints, a small test instance	The formulation below; three fixed instance sizes
3. Justify	The quantum algorithm and how the problem is represented in it	QAOA with a start state and mixer that keep exactly k tests
4. Implement	A meaningful quantum component in Qiskit	The quantum module in the repo, run on simulator and one real chip
5. Evaluate	A classical, exact or analytically verifiable reference	Brute force, an exact solver, the field officer's rule, and textbook genetics values
6. Reflect	Correctness, feasibility, limitations, trade-offs	Limitations and ethics sections of the report

Submission items, all four required:

- GitHub repository link with working, reproducible Qiskit code
- Report (.docx) with results, the classical or exact comparison and an honest limitations section
- Architecture diagram, included in the report
- 3-minute video pitch and demo, with link access set to "anyone with the link"

Submit through the submission form.

Dos and don'ts, as a final checklist:

- [ ] Instance is small and simulator-friendly
- [ ] The quantum part does real work in the solution
- [ ] README has setup steps a judge can follow
- [ ] Report says plainly what did not work
- [ ] Every link opens in a private browser window
- [ ] Problem stays inside the Healthcare domain
- [ ] All reused code and methods are credited
- [ ] Video is 3:00 or shorter
- [ ] No commits after the deadline

Domain fit. The brief's Healthcare examples include "allocation of limited medical equipment". This project allocates limited test kits and field-officer visits, so it sits inside the domain.

The problem

For one family visit with k test kits, choose the relatives to test so that the expected number of at-risk couples found is as large as possible.

Plain words. Thalassemia major appears when both parents are carriers. Screening teams start from a family that already has an affected child and test the relatives. Kits, lab capacity and visits are limited, so the team must choose. A couple's risk is only known when both partners are tested, so the best choice depends on pairs of people, not on individuals.

Why it matters here. Figures from the two sources this project leans on:

- About 5 percent of Pakistanis carry beta-thalassemia (NEJM, 2002).
- In that study 183 of 591 tested relatives in index families were carriers (31 percent), against 0 of 397 in control families.
- 17 of 214 married couples in those families were both carriers (8 percent).
- Punjab's prevention project covers all 36 districts with 48 field officers, and describes uptake as suboptimal (EJHG, 2021).

Islamabad's Compulsory Thalassemia Screening Act, 2025 makes this a legal duty there: institutions treating patients must trace and counsel their blood relatives (Friday Times, Aug 2026).

Who it serves. Field officers planning a home visit, and programme managers spreading kits across families.

Element	Definition
Decision variable	x_i = 1 if relative i is tested in this visit, otherwise 0
Pair weight w_ij	Probability that i and j are both carriers, for each couple that is married, engaged or likely to marry
Single weight p_i	Probability that i is a carrier
Relevance r_i	How much finding carrier i matters, for example higher for unmarried adults than for grandparents

Element	Definition
Objective	Expected at-risk couples found, plus a small reward for carriers found
Constraint	Exactly k people are tested
Optional	People who decline are fixed to 0; a cap on households visited

$$\max_{x \in \{0,1\}^n} \; \sum_{(i,j) \in C} w_{ij}\, x_i x_j \;+\; \lambda \sum_{i} r_i\, p_i\, x_i \qquad \text{subject to} \qquad \sum_{i} x_i = k$$

The first term is exact, not an approximation. It equals the expected number of couples where both partners are tested and both are carriers.

Where the weights come from. A pedigree engine computes p_i and w_ij from the family tree, the known affected child and the 5 percent population rate. It is checked against textbook values: parents of an affected child 1, an unaffected sibling 2/3, grandparents, aunts and uncles about 1/2, first cousins about 1/4.

Calibration target. Synthetic families should land near the NEJM figures above: about 31 percent carriers among relatives and about 8 percent of couples both carriers.

Instance	People	Kits k	Used for
S	10 to 12	3 to 4	Brute force, real-chip run, video demo
M	16 to 20	5 to 6	Main simulator results
L	40 to 60, several linked families	10 to 15	Exact classical solver; decomposition stretch goal

All instances are synthetic and seeded. No real patient data is used at any point.

Open questions for a practitioner, week 1:

- [ ] Are samples collected in one visit with results returned later? The model assumes the choice is made before any result is known. The EJHG paper does not describe this step.
- [ ] Is the real limit kits, lab slots or field-officer time?
- [ ] Which couples count: married, engaged, or likely matches inside the family?

Quantum approach in Qiskit

QAOA runs with a start state and mixer that can only output selections of exactly k people, so the kit limit needs no penalty term.

Problem piece	Quantum piece	Qiskit building block
Is relative i tested?	Qubit i	`QuantumCircuit`
Couple weight w_ij	ZZ rotation on qubits i and j	`rzz`, `SparsePauliOp`
Single weight	Z rotation on qubit i	`rz`
Exactly k tests	Dicke start state plus XY ring mixer	Custom circuit, `XXPlusYYGate`
Tuning the angles	Classical optimiser loop	SciPy COBYLA with the Estimator primitive
Reading out plans	Sampling the circuit	Sampler primitive, Aer, IBM Runtime

Two quantum variants are compared:

1. Standard QAOA with a penalty for using the wrong number of kits. This is the baseline.
2. Constrained QAOA with a Dicke start state and XY mixer. This is the main method. It follows Cook, Eidenbenz and Bärtschi, who report that Dicke states beat simple start states on a sibling problem.

Feasibility check, run on 12 Nov 2026 with Qiskit 2.6.2. One random toy instance: 10 people, 4 kits, 17 couple links, noise-free simulation.

Method	Quality score	Chance of the best plan
Random valid pick	0.43	0.5%
Constrained QAOA, depth 1	0.63	2.0%
Constrained QAOA, depth 2	0.69	3.8%
Constrained QAOA, depth 3	0.72	6.1%

Quality score is the expected value, scaled so the worst valid plan is 0 and the best is 1. Every sampled plan used exactly 4 kits. Three caveats apply: it is one instance, the simulation has no noise, and the start state was loaded directly instead of built from gates.

Known gap. Qiskit 2.6.2 has no built-in Dicke state. Bärtschi and Eidenbenz give a gate-level scheme with O(kn) gates and no helper qubits. Building it is a Days 11 to 15 task.

Classical references:

- Brute force over every valid set of k people, exact for S and M sizes
- An exact integer-programming solver for L size
- The field officer's rule: test the k people with the highest carrier probability
- A random valid pick
- Simulated annealing with swap moves

Extensions, in priority order:

1. **Noise.** Run on an offline model of a real IBM chip. Discard shots without exactly k ones and report the share that survives.
2. **Real chip.** One S-size run on IBM's free Open Plan, which allows up to 10 minutes of processor time per 28 days (IBM docs). Angles are trained on the simulator first.
3. **Trade-off curve.** Sweep the balance between couples found and carriers found, and report the plans no other plan beats on both. This borrows the idea of IBM's multi-objective QAOA paper.
4. **Linked families.** Split a large network into family clusters, solve each cluster for each kit count, and split the budget classically.

How we defend "why quantum":

- **Natural fit.** The objective is pairwise on yes/no decisions, which is QAOA's native form. The kit limit is enforced by the circuit's structure.

- **Hard in general.** With equal weights the problem is densest-k-subgraph, whose best known efficient algorithm only guarantees a factor of about n^(1/4) (ECCC TR16-195). That is a worst-case statement. Sparse family graphs may be much easier.
- **Measured, not assumed.** Every quantum number sits beside brute force, an exact solver and the field rule.
- **Known weak spot.** One family is small enough for an exact classical solver. We say so, and present the work as a method study on a new problem.

The hardness results for densest-k-subgraph rest on complexity assumptions that the ECCC page shows were revised. Read the current revision before quoting them.

One-sentence answer for judges: "We do not claim a speedup. We show a problem QAOA represents directly, a circuit that cannot break the kit limit, and an honest comparison against exact answers."

Rubric map

Each of the 15 rubric lines has a named piece of evidence, a place it appears and a day it is finished. The 100 points split as 20, 25, 25, 15 and 15.

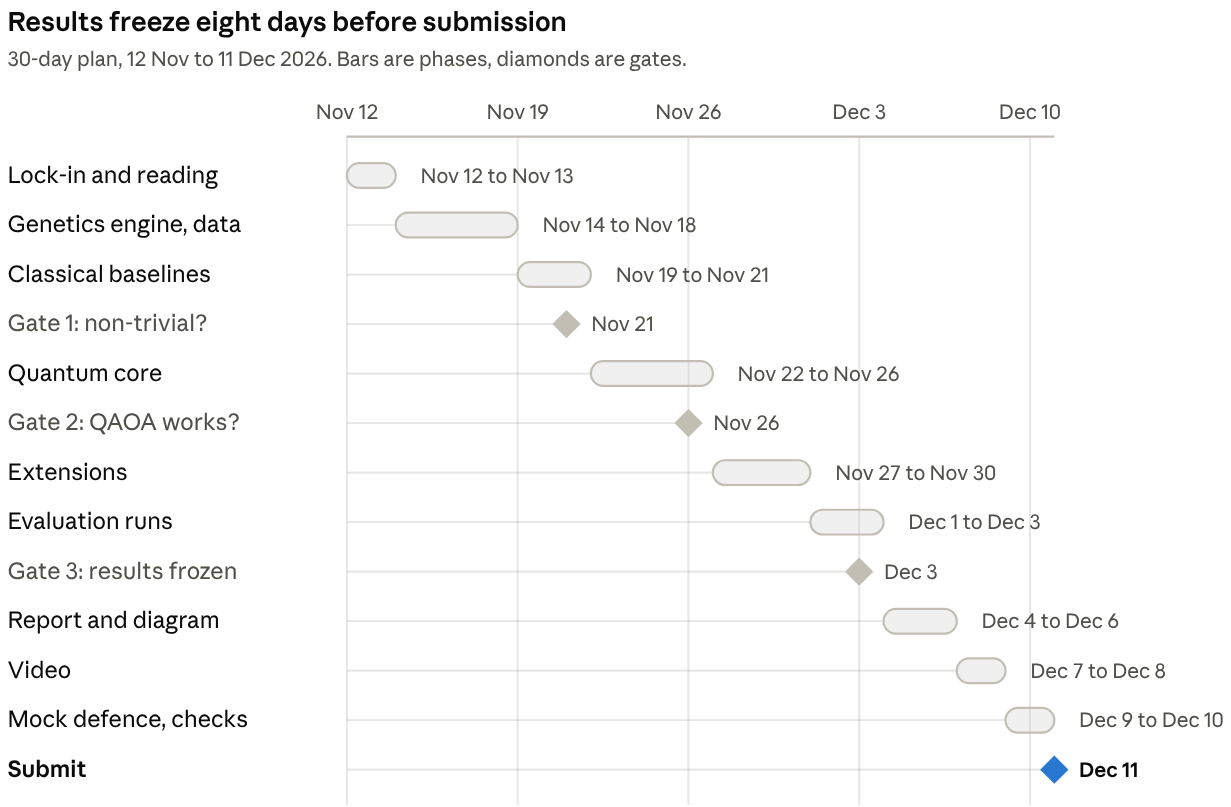
Rubric line	Points	Evidence we produce	Where it shows	Done by
A. Relevance within the domain	5	Sourced Pakistan figures; one practitioner conversation	Report section 1; video opening	Day 5
A. Clear optimisation formulation	5	Variable, objective and constraint table with the formula	Report section 2; README	Day 9
A. Scope and feasibility	5	Three fixed instance sizes; go/no-go result	Report section 2	Day 10
A. Measurable objectives and constraints	5	Expected at-risk couples found per k kits; exactly-k rule	Report section 2; results tables	Day 9
B. Correct mapping to a quantum representation	10	Mapping table; a test that circuit energy equals the objective for every input	Report section 3; tests folder	Day 13

Rubric line	Points	Evidence we produce	Where it shows	Done by
B. Suitability and justification of algorithm	10	Why QAOA; penalty against constrained comparison; hardness note with its caveat	Report section 3	Day 16
B. Quantum materially used; awareness of limits	5	The plan comes from measured circuits; comparison against random picks; limits list	Report sections 3 and 6	Day 22
C. Working Qiskit implementation	10	One-command demo on the simulator; one real-chip run	Repo; video demo	Day 19
C. Interpretable outputs	5	Family tree with chosen people highlighted; plain-language plan	Demo notebook; report figures	Day 21
C. Valid classical, exact or analytical baseline	5	Brute force, exact solver, field rule; textbook genetics checks	Report section 5	Day 22
C. Reproducibility	5	Pinned requirements, fixed seeds, saved results, fresh-clone test	README; repo	Day 29
D. Originality of the problem	10	Literature-check note; the cascade-screening framing	Report section 1	Day 2
D. Adaptation, modelling or technical extension	5	Pedigree-to-weights engine; constrained mixer; trade-off curve	Report sections 3 and 4	Day 19
E. Clarity of explanation	10	Storyboard, one architecture diagram, two rehearsals	Video; report	Day 27
E. Demo quality and technical answers	5	Recorded live demo; question-bank drill	Video; mock defence	Day 28

Sections B and C carry half the marks, so the quantum core and the evaluation runs get 12 of the 30 days.

30-day schedule

The 30 days run in nine phases with three gates. Results freeze on Day 22, so the last eight days go to the report, the video and checks.



30-day plan · 9 phases, 3 gates

Day 1 is Thursday 12 Nov and Day 30 is Friday 11 Dec. Each gate is a yes or no decision that the whole team makes together.

Days 1 to 2 (12 to 13 Nov): lock-in

- [ ] Confirm the submission date and the video rules with the organisers
- [ ] Put a name on each workstream; create the GitHub repo with a licence, README stub and pinned requirements
- [ ] Run the novelty search on arXiv and Google Scholar; write a half-page note on what exists

- [ ] Read the two source papers and the three method papers
- [ ] Write the one-page problem statement
- [ ] Contact one thalassaemia centre or haematologist with the three open questions

Days 3 to 7 (14 to 18 Nov): genetics engine and data

- [ ] Build the family generator: generations, family size, share of cousin marriages
- [ ] Build the probability engine for p_i and w_ij, given the affected child
- [ ] Test it against the textbook values
- [ ] Calibrate to the NEJM figures
- [ ] Freeze seeded S, M and L instance files in the repo
- [ ] Draft the ethics note

Days 8 to 10 (19 to 21 Nov): formulation and classical baselines

- [ ] Code the objective and the exactly-k rule, with tests
- [ ] Implement brute force, the field rule, random pick and simulated annealing
- [ ] Add the exact solver for L size
- [ ] Draw the family tree output with the chosen people highlighted
- [ ] Gate 1, Day 10: on 50 seeded instances, how often does the field rule miss the best plan, and by how much?

Days 11 to 15 (22 to 26 Nov): quantum core

- [ ] Build the cost circuit; test that its energy equals the objective for every input on a small case
- [ ] Run penalty QAOA on S: angles, sampling, metrics
- [ ] Build the Dicke state from gates, and the XY ring mixer
- [ ] Run constrained QAOA on S and M at depths 1 to 3
- [ ] Compare penalty against constrained: quality, share of valid shots, circuit depth
- [ ] Gate 2, Day 15: constrained QAOA beats random picks on S and M, and every noise-free shot has exactly k ones

Days 16 to 19 (27 to 30 Nov): extensions

- [ ] Noise simulation on an offline chip model, with the share of discarded shots
- [ ] Trade-off curve from the weight sweep

- [ ] One real-chip run on S with simulator-trained angles (Could tier)
- [ ] Linked-family decomposition on L (Could tier)

Days 20 to 22 (1 to 3 Dec): evaluation runs

- [ ] Run the full matrix: sizes S, M and L; depths 1 to 3; at least 20 seeds
- [ ] Save raw results to files; produce every figure from a script
- [ ] Write down what did not work, with numbers
- [ ] Gate 3, Day 22: results are frozen; no new experiments after this

Days 23 to 25 (4 to 6 Dec): report, diagram, README

- [ ] Write the report to the outline below and export it as .docx
- [ ] Put the final architecture diagram inside the report
- [ ] README: setup, one-command demo, expected output, credits
- [ ] A teammate who did not write a section reviews it

Days 26 to 27 (7 to 8 Dec): video

- [ ] Write the script to the storyboard; rehearse twice with a timer
- [ ] Record the screen demo and voice; cut to 2:45
- [ ] Upload, set access to anyone with the link, and test in a private window

Days 28 to 30 (9 to 11 Dec): defence, checks, submit

- [ ] Day 28: mock defence with the question bank; fix weak answers
- [ ] Day 29: fresh clone on a different machine, following the README only
- [ ] Day 29: final commit and a release tag; no commits after this
- [ ] Day 30: open every link in a private window, submit the form, and keep a screenshot of the confirmation

Workstreams, scope and risks

Three workstreams run in parallel, and scope is tiered so that a slip cuts extras, never deliverables.

Workstream	Owns	First hard output
Genetics and data	Family generator, probability engine, instance files, ethics note	Validated engine by Day 7
Classical and evaluation	Baselines, experiment runner, plots, statistics	Gate 1 result on Day 10
Quantum	Qiskit circuits, Dicke state, mixers, noise and real-chip runs	Constrained QAOA on the S instance by Day 15
Delivery, shared with one named lead	Repo hygiene, report, diagram, video, submission	Fresh-clone test on Day 29

Put a name on each row on Day 1. With two people, merge the first two rows.

Tier	Items	Rule
Must	Pedigree engine; formulation; penalty QAOA on the simulator; brute force and field rule; report; diagram; video; README	Never cut
Should	Dicke state with XY mixer; exact solver; noise simulation; trade-off curve	Cut only if unfinished on Day 19
Could	Real-chip run; linked-family decomposition; a third objective	Cut first

Risk	Signal	Response
Someone has already published this	The Day 1 to 2 search finds quantum work on cascade screening	Cite it and move the claim to the part that is new
The problem is trivial	At Gate 1 the field rule matches the best plan on most instances	Add realistic couple weights and a household cap; if still trivial, say so and lead with the trade-off curve
The quantum defence is thin	A judge asks why not an exact solver	Use the prepared answer; show results across S, M and L

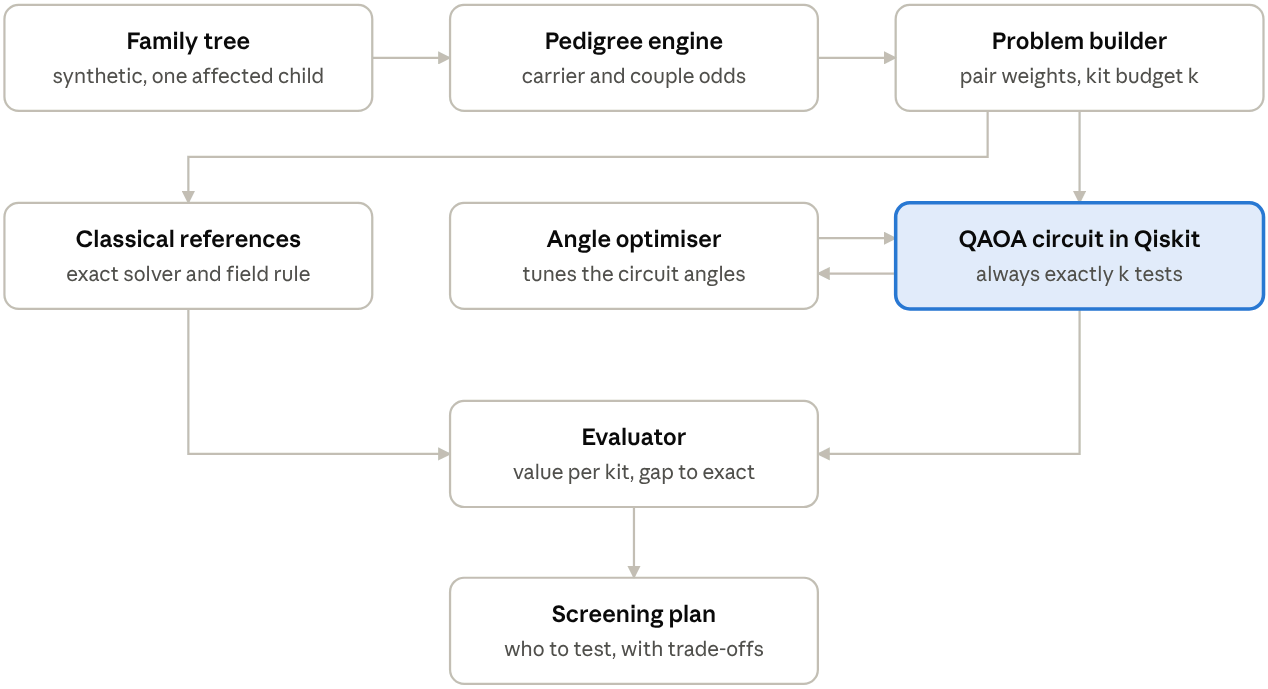
Risk	Signal	Response
The Dicke circuit is harder than expected	Not working by Day 13	Load the start state directly on the simulator, say so, and keep penalty QAOA as the fallback
No real-chip result	Queue or errors by Day 19	Use the offline noisy chip model and report that honestly
Ethics objection	A reviewer says the tool ranks people by genetic risk	Synthetic data only; consent is a model input; the tool supports counsellors and publishes no individual scores
Time slips	A gate is missed by more than two days	Drop the Could tier, then the Should tier
The deadline is earlier than assumed	Organisers confirm on Day 1	Deliver the Must tier only

Deliverable blueprints

The four submission items are specified here, so the last week is assembly and not design.

Architecture diagram

The quantum circuit proposes plans; classical code builds, tunes and checks



system architecture · 8 parts

Classical code prepares the weights, tunes the circuit and checks the answers. The quantum circuit is the step that proposes screening plans. Redraw this for the report once the modules exist.

Repository layout

```
kinship-qaoa/
  README.md              setup, one-command demo, credits
  requirements.txt       pinned versions
  LICENSE
  data/instances/        seeded S, M and L families
  kinship/
    pedigree.py          family generator
    genetics.py          carrier and couple probabilities
    problem.py           objective and exactly-k rule
    classical.py         brute force, exact solver, field rule, annealing
    quantum/
      circuits.py        cost layer, Dicke state, XY mixer
```

```
    qaoa.py            angle tuning and sampling
    hardware.py        noisy model and real-chip runs
  evaluate.py          metrics and experiment runner
  plots.py             figures
notebooks/demo.ipynb   the demo shown in the video
results/               raw outputs and figures
tests/                 genetics values, objective, circuit checks
report/                report.docx and diagram
```

Report outline (.docx, aim for 8 to 10 pages)

Section	Content	Brief step
1. Problem and users	Pakistan context, who it serves, problem statement	Identify
2. Formulation	Variables, objective, constraint, test instances	Formulate
3. Quantum method	Mapping, algorithm choice, why a constrained mixer	Justify
4. Architecture and implementation	The diagram, modules, how to run	Implement
5. Results	QAOA against the references, noise, real chip, trade-off curve	Evaluate
6. Limitations and what did not work	The list below, with numbers	Reflect
7. Ethics	Consent, stigma, synthetic data	Reflect
8. Credits and references	Papers, libraries, reused code	

Limitations to state up front:

- The model covers one visit; real programmes may test step by step.
- Families are synthetic and calibrated to one study from 2002.
- The test is assumed perfect and the genetics is one gene with standard inheritance.
- At these sizes an exact classical solver wins.
- Results are noise-free unless stated; only the smallest instance runs on a real chip.
- Worst-case hardness results do not automatically apply to family graphs.

Video storyboard (limit 3:00, target 2:45)

Time	On screen	What is said
0:00 to 0:25	A family tree with one affected child	One visit, k kits: whom do we test?
0:25 to 0:50	Couple links and the formula	The value sits on couples, so this is a pairwise problem
0:50 to 1:20	Architecture diagram, then the circuit	Why QAOA, and how the circuit keeps exactly k tests
1:20 to 2:10	Live demo: one command, chosen people light up, results table	QAOA against the exact answer and the field rule
2:10 to 2:35	Limitations slide	What did not work; no speedup is claimed
2:35 to 2:45	Repo link and team names	Close

Judge question bank

Every team member should be able to answer all eleven questions without notes by Day 28.

Why quantum, when a classical solver handles this size? At our sizes it does, and we say so. The project shows a problem whose structure QAOA represents directly and measures how close it gets to exact answers.

Did quantum beat classical? No. Brute force finds the best plan instantly on our instances. We report the gap between QAOA and that exact answer.

What is new here? The problem framing and the weights derived from genetics. The constrained QAOA method is borrowed from published work and credited.

Why not simply test the closest relatives? Because the value sits on couples, not individuals. Gate 1 measures how often that simple rule misses the best plan.

Why not test one partner first and the other only if positive? That is sensible when results return quickly. Our model covers the single-visit case, and the report lists step-by-step testing as a limitation and a next step.

How are the probabilities computed? By Mendel's rules on the family tree, given the affected child. The engine is checked against textbook values before any optimisation runs.

Is the data real? No. Families are synthetic and calibrated to published figures. No patient data is used.

What keeps the answer at exactly k tests? The start state has exactly k ones and the mixer only swaps a one with a zero. On noisy hardware some shots break this, so we discard them and report how many.

What happens on a real chip? Noise lowers quality, and building the start state from gates adds depth. We run only the smallest instance there and show the drop.

How does it scale? One qubit per relative. Simulators reach roughly 25 to 30 people. Beyond that the plan is to split linked families into clusters.

Is ranking relatives by genetic risk dangerous? It can be. The tool uses synthetic data, treats consent as an input, and is meant to support a counsellor's plan, not to label people.

Sources and credits

The brief says reused code must be credited, so the README and the report both carry the credit list below.

Credit in the README and the report:

- [ ] Constrained QAOA design with a Dicke start state and XY mixer: Cook, Eidenbenz and Bärtschi
- [ ] Dicke state circuit: Bärtschi and Eidenbenz
- [ ] Trade-off curve idea: Kotil and co-authors, Quantum Approximate Multi-Objective Optimization
- [ ] Qiskit, Qiskit Aer, Qiskit IBM Runtime, SciPy and NumPy, with version numbers
- [ ] Any tutorial or open-source code reused, with its link
- [ ] Ask the organisers whether AI assistance must be disclosed, and disclose it if in doubt

Pages opened for this plan:

- Screening extended families for genetic hemoglobin disorders in Pakistan, NEJM 2002
- Cascade screening for beta-thalassaemia in Pakistan, EJHG 2021
- Why Pakistan must stop treating thalassemia as fate, Friday Times, Aug 2026

- The Quantum Alternating Operator Ansatz on Maximum k-Vertex Cover
- Deterministic Preparation of Dicke States
- Quantum Approximate Multi-Objective Optimization
- Hardness of approximating Densest k-Subgraph, ECCC TR16-195
- IBM Quantum plans overview

Reading list for Days 1 to 2, found in search but not opened for this plan:

- Grover Mixers for QAOA, an alternative constrained mixer
- A Divide-and-Conquer Approach to Dicke State Preparation, which targets IBM's chip layout
- An O(n^1/4) approximation for Densest k-Subgraph

The novelty check so far is one web search per topic, which found no quantum work on cascade screening. A proper search on arXiv and Google Scholar is the first task on Day 1.